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$$w_{rev} = -\int_{V_i}^{V_f} P_{ext}\, dV$$

In a compression process, P_{ext} the external pressure is always greater than the pressure of the system.

i.e $P_{ext} = (P_{int} + dP)$.

In an expansion process, the external pressure is always less than the pressure of the system

i.e. $P_{ext} = (P_{int} - dP)$.

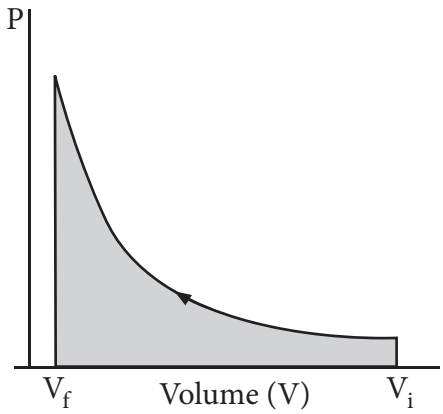


Figure 7.4

When pressure is not constant and changes in infinitesimally small steps (reversible conditions) during compression from V_i to V_f , the P-V plot looks like in fig 7.4 Work done on the gas is represented by the shaded area.

In general case we can write,

$P_{ext} = (P_{int} \pm dP)$. Such processes are called reversible processes. For a compression process work can be related to internal pressure of the system under reversible conditions by writing equation

$$w_{rev} = -\int_{V_i}^{V_f} P_{int}\, dV$$

For a given system with an ideal gas

$$P_{int}\, V = nRT$$

$$P_{int} = \frac{nRT}{V}$$

$$w_{rev} = -\int_{V_i}^{V_f} \frac{nRT}{V}\, dV$$

$$w_{rev} = -nRT \int_{V_i}^{V_f} \left(\frac{dV}{V}\right)$$

$$w_{rev} = -nRT \ln\left(\frac{V_f}{V_i}\right)$$

$$w_{rev} = -2.303\, nRT \log\left(\frac{V_f}{V_i}\right) \text{ ------ } 7.6$$

If $V_f > V_i$ (expansion), the sign of work done by the process is negative.

If $V_f < V_i$ (compression) the sign of work done on the process is positive.

Table: 7.3 Summary of sign conventions

1.	If heat is absorbed by the system	:	$+q$
2.	If heat is evolved by the system	:	$-q$
3.	work is done by the system	:	$-w$
4.	work is done on the system	:	$+w$

7.3 Zeroth law of thermodynamics:

The zeroth law of thermodynamics, also known as the law of thermal equilibrium, was put forward much after the establishment of the first and second laws of thermodynamics. It is placed